AWS IoT Core vs Azure IoT Hub

A Comprehensive Comparison

Agenda

1. Limitations Comparison
2. Pricing Comparison
3. Security Features Comparison
4. Overall Assessment & Recommendations

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1. Limitations Comparison

Message Processing Capacity

Feature	AWS IoT Core	Azure IoT Hub
Max messages per connection	100 msgs/sec	Tier-based (S1: 100/sec, S2: 120/sec, S3: 6,000/sec)
Max bandwidth per connection	512 KB/sec	Tier-based
Max message size	128 KB	256 KB (device-to-cloud), 64 KB (cloud-to-device)
Max concurrent connections	Unlimited	Tier-based
New connection rate	3,000/sec per account	Tier-based

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1. Limitations Comparison (cont.)

Key Differences in Message Processing

- **AWS IoT Core**: Fixed limit of 100 messages/second per connection (cannot be increased)
- **Azure IoT Hub**: Higher capacity in S3 tier (6,000 messages/second per unit)
- **Bottleneck Risk**: AWS may create bottlenecks in high-volume scenarios
- **Scalability**: Azure offers better vertical scaling through tier selection

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1. Limitations Comparison (cont.)

Protocol Support

Protocol	AWS IoT Core	Azure IoT Hub
MQTT	✓	✓
MQTT over WebSocket	✓	✓
HTTPS	✓ (publish only)	✓
AMQP		✓
AMQP over WebSocket		✓

Key Insight: Azure offers more protocol options, especially AMQP which is valuable in enterprise scenarios

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2. Pricing Comparison

AWS IoT Core Pricing Structure

- **Component-based pricing** with 5 main elements:

- **Connectivity**: $0.08/1,000,000 connection minutes

- ~$0.042/year per device for 24/7 connectivity

- **Messaging**: ~$1 per million messages

- Metered in 5 KB increments
- **Device Shadow & Registry**: Operations that store device state
- Metered in 1 KB increments
- **Rules Engine**: Rules for transforming and routing data
- Metered in 5 KB increments
- **Device Location**: Location solutions

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2. Pricing Comparison (cont.)

Azure IoT Hub Pricing Structure

- **Tier-based pricing** with 3 main tiers:
- **S1 Tier**:
- 400,000 messages/day per unit
- $25/month per unit
- **S2 Tier**:
- 6,000,000 messages/day per unit
- $250/month per unit
- **S3 Tier**:
- 300,000,000 messages/day per unit
- $2,500/month per unit
- Multiple units can be purchased for each tier

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2. Pricing Comparison (cont.)

Cost Comparison for Your Scenario

Scenario: Year 1: 50,000 devices, Year 2: 100,000 devices, 200 messages/device/day

AWS IoT Core (2-year total: $17,250)

- Year 1: $5,750 ($2,100 connectivity + $3,650 messaging)
- Year 2: $11,500 ($4,200 connectivity + $7,300 messaging)

Azure IoT Hub - S2 Tier (2-year total: $18,000)

- Year 1: $6,000 (2 units × $250/month × 12 months)
- Year 2: $12,000 (4 units × $250/month × 12 months)

Key Insight: Similar total costs over 2 years

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3. Security Features Comparison

AWS IoT Device Defender vs Azure Defender for IoT

Feature	AWS IoT Device Defender	Azure Defender for IoT
Device auditing	✓	✓
Continuous monitoring	✓	✓
Anomaly detection	✓ (ML-based)	✓ (ML-based)
Vulnerability management	✓	✓
OT/ICS device support	Limited	Comprehensive
Network traffic analysis	Limited	Comprehensive
Deployment model	Cloud only	Cloud and on-premises

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3. Security Features Comparison (cont.)

AWS IoT Device Defender

- **Auditing**: Checks security configuration against best practices
- **Detection**: ML models to detect abnormal device behaviors
- **Metrics**: Monitors 13 different metrics (cloud-side and device-side)
- **Automatic Learning**: Uses 14-day trailing data to learn behavior patterns
- **Integration**: Tightly integrated with AWS IoT Core

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3. Security Features Comparison (cont.)

Azure Defender for IoT

- **Comprehensive Coverage**: Supports IoT and OT/ICS devices
- **Network Monitoring**: Passive monitoring with deep protocol analysis
- **Threat Intelligence**: Integrated with Microsoft's threat intelligence
- **Deployment Flexibility**: Both cloud and on-premises options
- **Zero Touch**: Can monitor existing devices without agents
- **Industrial Focus**: Stronger in industrial and critical infrastructure scenarios

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4. Overall Assessment

Advantages of AWS IoT Core

1. **Flexibility**: Component-based, pay-as-you-go pricing
2. **AWS Ecosystem**: Seamless integration with AWS services
3. **High Scalability**: Practically unlimited device connections
4. **Simple Start**: Lower complexity for initial implementation

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4. Overall Assessment (cont.)

Advantages of Azure IoT Hub

1. **Higher Message Throughput**: 6,000 msgs/sec/unit in S3 tier
2. **Protocol Diversity**: AMQP support for enterprise scenarios
3. **Larger Messages**: 256 KB device-to-cloud (vs 128 KB in AWS)
4. **Comprehensive Security**: Better for industrial IoT scenarios

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4. Overall Assessment (cont.)

Recommendations for Your Scenario

For 50,000-100,000 devices sending 200 messages/day each:

1. **Message Distribution**: If messages are evenly distributed, both platforms work well; if concentrated in time periods, Azure may be better

2. **Cost Considerations**: Similar total costs; AWS more granular, Azure more predictable
3. **Security Requirements**: For industrial IoT or comprehensive security, Azure has an edge
4. **Existing Ecosystem**: Choose based on your current cloud platform for better integration

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Thank You!

Questions?